1 Background

The Baker–Wurgler sentiment index (hereafter the BW index; Baker and Wurgler (2006)) is constructed by aggregating six market-based sentiment proxies[1] and orthogonalizing these proxies with respect to a set of macroeconomic controls.

In the original implementation, these macro variables were transformed into year-over-year growth rates before entering the orthogonalization regressions. During the COVID-19 period, however, this transformation produced mechanically extreme values because the denominator of the year-over-year ratio corresponded to unusually depressed economic conditions in early 2020. These base-effect-driven spikes subsequently distorted the orthogonalization step, generating a temporary divergence between the raw sentiment index and its macro-orthogonalized version.

In response, we revise the macro-processing procedure so that the sentiment orthogonalization is based on *detrended* macro series rather than year-over-year differences. This section outlines the distinctions between the original and revised implementations and compares the resulting series. The evidence suggests that the detrended specification yields a macro-orthogonalized BW index that more closely tracks the raw sentiment series, particularly during periods of post–COVID rebound.

2 What the Original Code Does

In the original Stata code, each macroeconomic variable X_t was converted into a year-over-year growth rate. For industrial production and employment, the transformation took the

[1] Although the original paper incorporated six sentiment variables, the turnover measure was subsequently dropped, leaving five proxies in the current implementation.

form

$$g_t = \frac{X_t}{X_{t-12}} - 1.$$

For the three consumption measures (durables, nondurables, and services), the same calculation was applied after deflating by the CPI:

$$g_t = \left(\frac{X_t/\text{CPI}_t}{X_{t-12}/\text{CPI}_{t-12}} \right) - 1.$$

The code also generated lagged values g_{t-12}, because the macro-orthogonalization step regresses each sentiment proxy (and its lagged variant) on both current and lagged macro controls. Conceptually, the orthogonalization regression takes the form

$$\text{Proxy}_t = \alpha + \beta_0 g_t + \beta_1 g_{t-12} + \beta_2 \text{Recess}_{t-1} + \varepsilon_t.$$

A key implication of this setup is that all macro controls depend directly on the value of the series exactly twelve months earlier. When X_{t-12} is unusually low—as it was during the early COVID-19 lockdowns—the resulting growth rates become artificially large.

[INSERT FIGURE 1 HERE]

These extreme observations then enter directly into the orthogonalization regressions, distorting the residuals and producing the divergence between the raw sentiment index and its macro-orthogonalized counterpart. Figure 1 illustrates this pattern: the macro-orthogonalized BW index (red solid line) deviates noticeably from the raw BW index (blue solid line) between March and July 2021, reflecting the impact of the anomalous macroeconomic values during this period.

3 What the New Code (Detrending) Does

The revised procedure replaces year-over-year growth rates with detrended macro deviations, eliminating the dependence on a single lagged denominator X_{t-12} and mitigating the COVID-era base effects. The updated approach proceeds through the following steps:

- **Log transformation and deflation.** The consumption categories are first deflated using the CPI, and all macro series are converted to logs,

  $$x_t = \ln(X_t),$$

  which stabilizes variance and preserves proportional interpretation.

- **Trend extraction.** A smooth trend component x_t^{trend} is obtained using a 12-month moving-average filter.[2]

- **Detrended deviations.** The cyclical macro component is defined as the percent deviation of the logged series from its estimated trend:

  $$g_t = \exp\big(x_t - x_t^{\text{trend}}\big) - 1.$$

  This transformation captures short-run macroeconomic fluctuations while smoothing over extreme swings caused by unusually low base levels, such as those experienced in early 2020.

- **Lag construction.** To maintain compatibility with the original BW orthogonalization framework, lagged detrended deviations are constructed analogously to the old

[2] Although a Hodrick–Prescott (HP) filter is a common alternative, concerns raised in Hamilton (2018) regarding the statistical properties of the HP filter motivated the choice of the moving-average specification for the baseline implementation.

growth-rate specification:

$$g_{t-12}.$$

These lagged series enter the orthogonalization regressions in the same manner as before.

Overall, the detrending-based transformation produces macro controls that are smoother, less sensitive to temporary denominator effects, and more robust during periods of unusual macroeconomic volatility.

Summary of the Detrending Results

Figure 2 shows the results. The macro-orthogonalized BW index constructed from the detrended macro series (red solid line) now closely tracks the raw BW index (blue solid line), eliminating the temporary divergence present in the original implementation. The revised macro-orthogonalized series also remains highly consistent with the original macro-orthogonalized series: the correlation between the two exceeds 0.99.[3] Overall, the detrended specification preserves the spirit of the BW macro-orthogonalization while providing substantially improved robustness during periods of heightened macroeconomic volatility.

[3]Specifically, the correlation is 0.9969 over the pre-COVID period (1965:07–2019:12) and 0.9947 over the full sample (1965:07–2023:12).

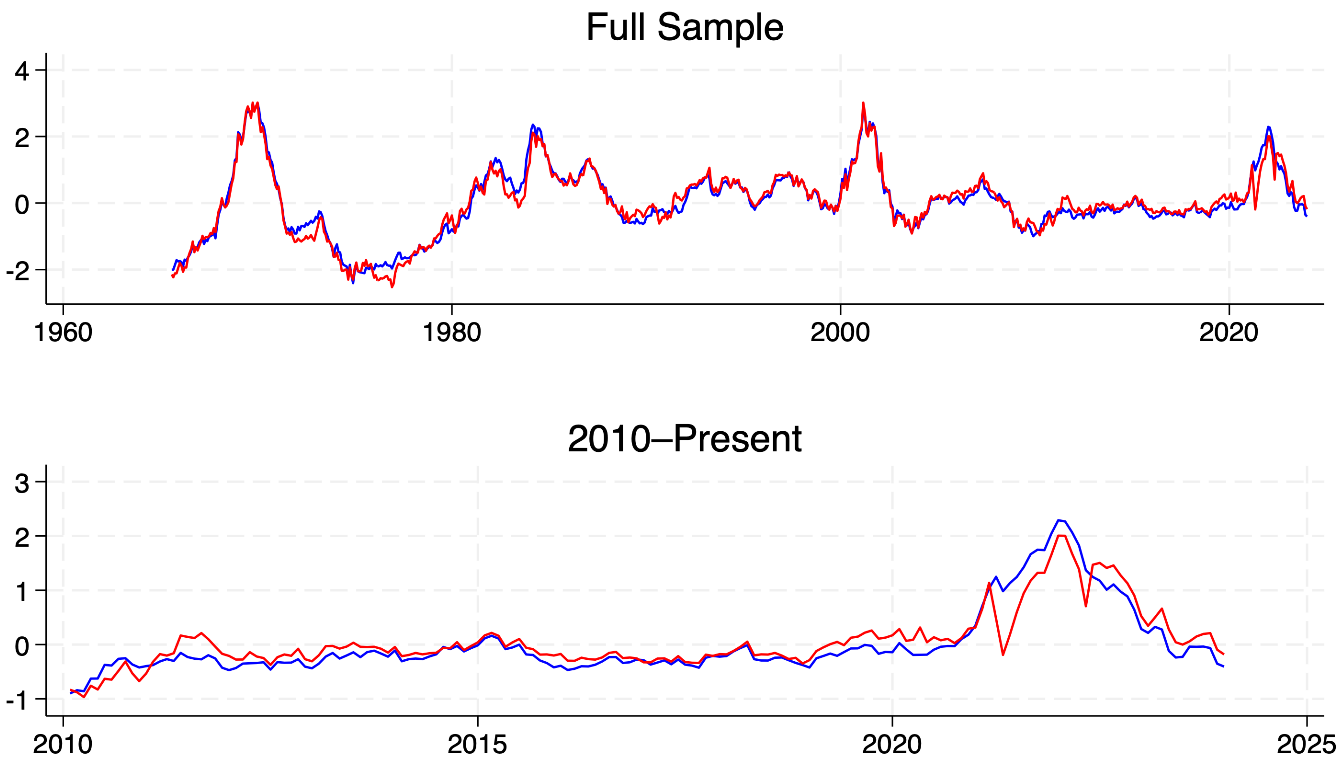
References

Baker, M. and Wurgler, J.: 2006, Investor Sentiment and the Cross-Section of Stock Returns, *Journal of Finance* **61**(4), 1645–1680.

Hamilton, J. D.: 2018, Why You Should Never Use the Hodrick-Prescott Filter, *Review of Economics and Statistics* **100**(5), 831–843.

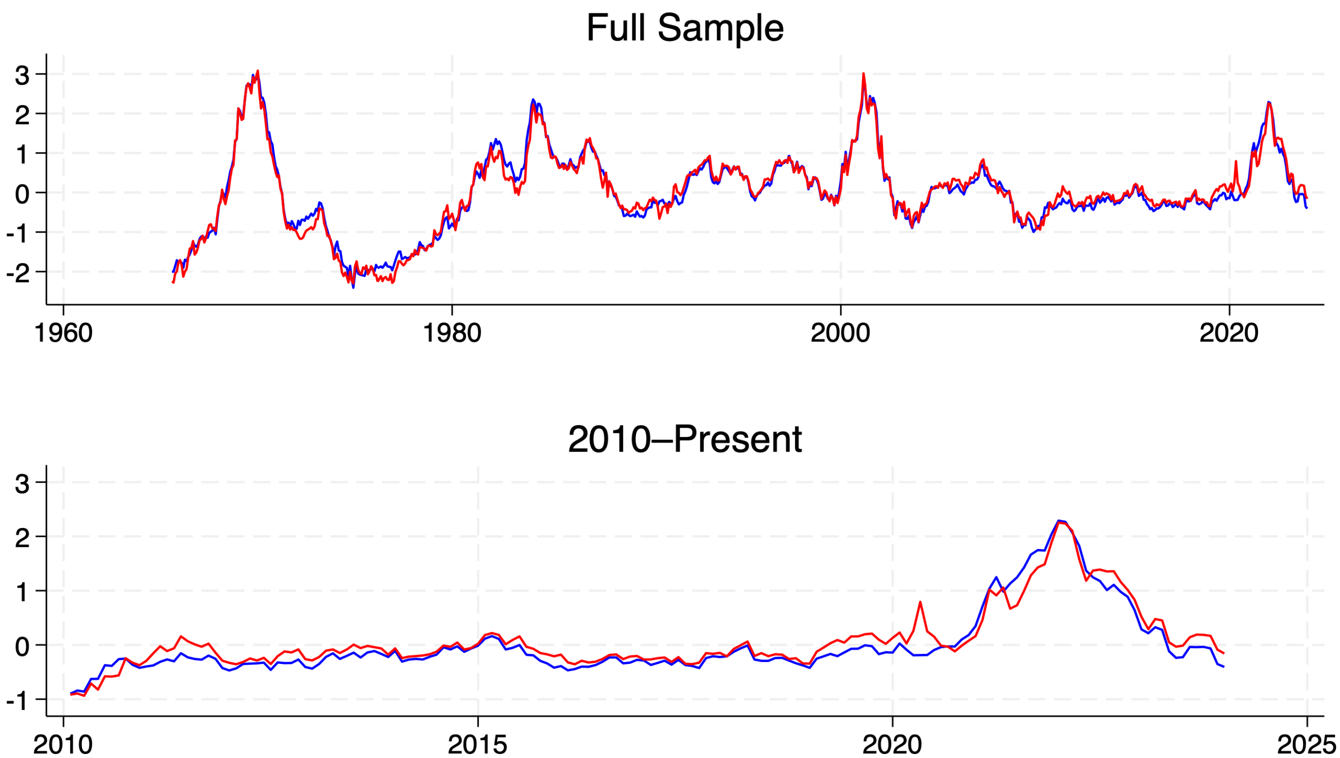
List of Figures

Figure 1: **BW Index Time-Series (Original STATA Code)**



Note: The figure plots the Baker–Wurgler Investor Sentiment Index using the original STATA implementation. The top panel shows the full sample (1965:07–2023:12) and the bottom panel shows the post-2010 period. The blue line denotes the raw sentiment index, while the red line shows the macro-orthogonalized version.

Figure 2: **BW Index Time-Series (NEW STATA Code)**



Note: The figure plots the Baker–Wurgler Investor Sentiment Index using the revised Stata implementation, which employs detrended macro series rather than year-over-year growth rates. The top panel shows the full sample (1965:07–2023:12) and the bottom panel shows the post-2010 period. The blue line denotes the raw sentiment index, while the red line shows the macro-orthogonalized version.